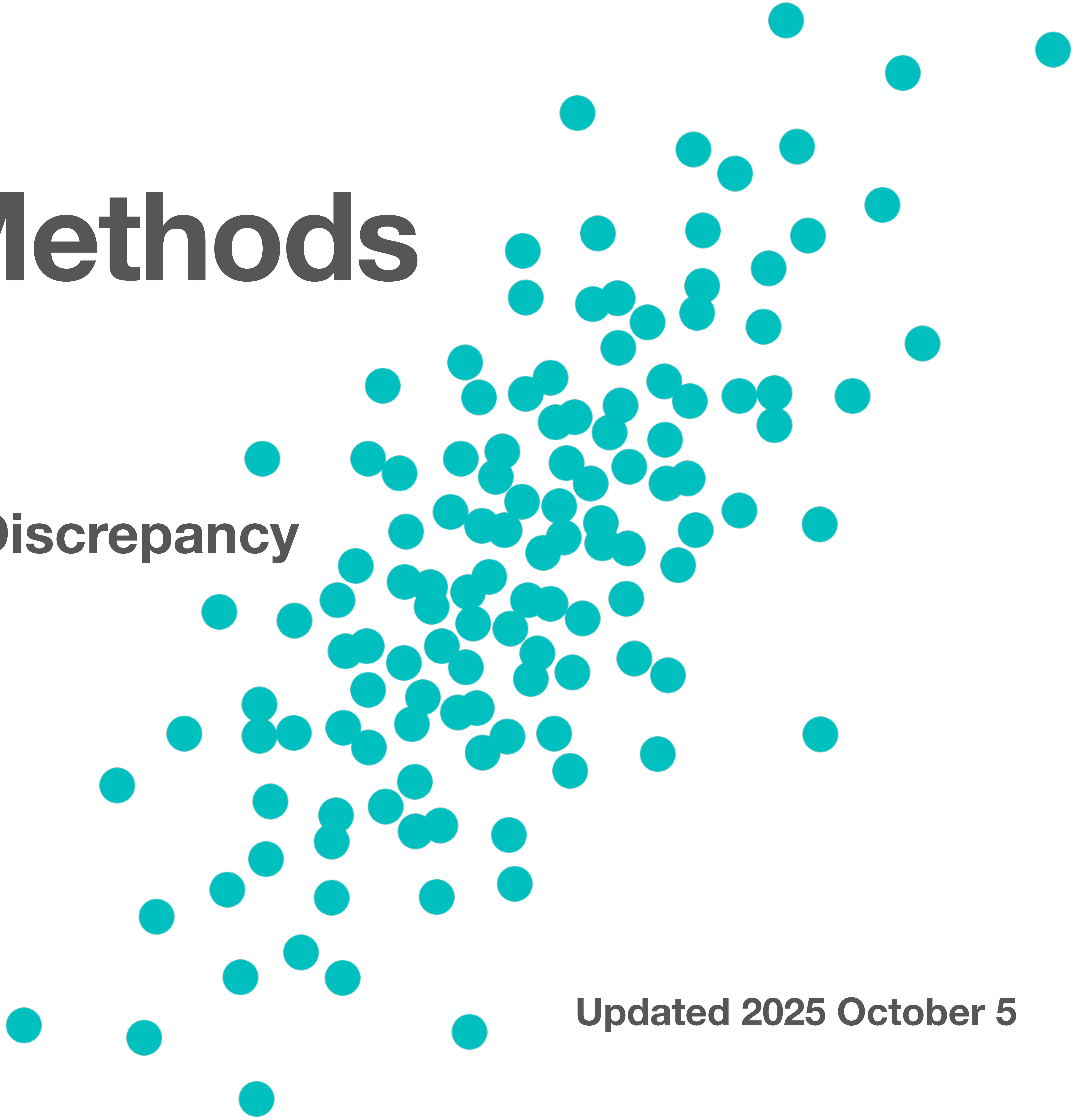
Monte Carlo Methods

Introduction
Generating Samples
Markov Chain Monte Carlo + Discrepancy
Improving Efficiency
Selected Topics

Git website and repository
Canvas



Fred Hickernell, Fall 2025

Updated 2025 October 5

Markov Chain Monte Carlo (MCMC) + Discrepancy

Owen, Chapters 11– 12

Project Selection due Oct 1

Assignment 3 due Oct 17

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025

Rate your confidence heading into Test 1

- Go to menti.com
- Use code 4923 2283

How do we generate samples when the distribution is complicated?

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  - An unnormalized target density
  - May reject a lot of samples
- Markov chain Monte Carlo (MCMC) moves samples toward areas of higher target density

Markov Chains

Let $\mathscr{X}$ be a state space for a sequence of random variables. Markov chains follow a probability law satisfying

$$\mathbb{P}(\boldsymbol{X}_{n+1} \in \mathscr{A} \mid \boldsymbol{X}_n = \boldsymbol{x}_n, \ldots, \boldsymbol{X}_0 = x_0) = \mathbb{P}(\boldsymbol{X}_{n+1} \in \mathscr{A} \mid \boldsymbol{X}_n = \boldsymbol{x}_n)$$

where $\mathscr{A}$ is some (measurable) subset of $\mathscr{X}$. This means that where you go next only depends on where you are now, not how you got there.

We use these to

- Generate $\boldsymbol{X}_0, \boldsymbol{X}_1, \ldots$ whose probability distribution converges to the target
- Study queuing systems

We are often interested in the asymptotic behavior of the $\boldsymbol{X}_i$

Metropolis–Hastings algorithm

Suppose that

- You really want to sample X with known (unnormalized) target density ϱ
- You have a proposal density $\varrho_{\mathrm{new|old}}$ to select the next point starting from where you are

```
Given X[0]
For i = 0 to n-1
  Generate Z ~ ϱ_new|old( · |X_i), U ~ 𝒰[0,1]
  If U ≤ min(1, ϱ(Z)ϱ_new|old(X_i|Z) / ϱ(X_i)ϱ_new|old(Z|X_i)) = min(1, joint density of first Z then X_i / joint density of first X_i then Z)
     X[i+1]←Z
  Else
     X[i+1]←X[i]
```

$$U \le \min\left(1, \frac{\varrho(Z)\varrho_{\mathrm{new|old}}(X_i|Z)}{\varrho(X_i)\varrho_{\mathrm{new|old}}(Z|X_i)}\right) = \min\left(1, \frac{\texttt{joint density of first } Z \texttt{ then } X_i}{\texttt{joint density of first } X_i \texttt{ then } Z}\right)$$

Note that you always accept either the new or the old

Metropolis algorithm

If $\varrho_{\mathrm{new|old}}(x \,|\, z) = \varrho_{\mathrm{new|old}}(z \,|\, x)$, then this can be simplified

Metropolis sampling proceeds as follows

Given X[0]

For i = 0 to n−1

  Generate Z $\sim \varrho_{\mathrm{new|old}}(\,\cdot\,|X_i)$, U $\sim \mathscr{U}[0,1]$

  If $U \le \min\left(1, \dfrac{\varrho(Z)}{\varrho(X_i)}\right)$

    X[i+1]←Z

  Else

    X[i+1]←X[i]

Advantages of and Challenges in MCMC

- Samples from complicated, unnormalized densities
- Easy to implement
- Proposal density and starting point need tuning to ensure that
    - Explore all the sample space and not miss out
    - Exploit sampling in regions where the density is large

How do we measure quality of MCMC

Short answer: We cannot easily

Longer answer: Given a symmetric, positive definite kernel, $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, the discrepancy, sometimes called the maximum mean discrepancy between two empirical distributions is defined by

$$D^2\left(\{\boldsymbol{x}_i\}_{i=0}^{m-1}, \{\boldsymbol{z}_i\}_{i=0}^{n-1}; K\right) = \frac{1}{m^2}\sum_{i,j=0}^{m-1} K(\boldsymbol{x}_i, \boldsymbol{x}_j) - \frac{2}{mn}\sum_{i,j=0}^{m-1,n-1} K(\boldsymbol{x}_i, \boldsymbol{z}_j) + \frac{1}{n^2}\sum_{i,j=0}^{n-1} K(\boldsymbol{z}_i, \boldsymbol{z}_j)$$

Requires $\mathcal{O}\left(\max(m,n)^2\right)$ operations

What is discrepancy?

The discrepancy measures

- The difference between two probability distributions
- The worst-case error in approximating an expectation/integral $\mu = \mathbb{E}[f(\boldsymbol{X})]$ by a sample mean when f is in the unit ball of a Hilbert space with reproducing kernel K
- The root-mean square error in approximating an expectation/integral $\mu = \mathbb{E}[f(\boldsymbol{X})]$ by a sample mean when f is an instance of a Gaussian process with covariance kernel K

Discrepancies from kernels were popularized by [Hickernell 1998] and [Gretton et al. 2012]

Symmetric, Positive Definite Kernels

- A square matrix, K, is symmetric and positive definite iff

$$\mathsf{K}^\mathsf{T} = \mathsf{K} \quad \text{and} \quad \boldsymbol{c}^T\mathsf{K}\boldsymbol{c} > 0 \text{ for all } \boldsymbol{c} \neq \boldsymbol{0}$$

- A kernel, $K : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$, is symmetric and positive definite iff

$$\mathsf{K} := \big(K(\boldsymbol{x}_i, \boldsymbol{x}_j)\big)_{i,j=0}^{n-1}, \qquad \text{the Gram matrix, is symmetric and positive definite}$$

for all $n \in \mathbb{N}$ and $\{\boldsymbol{x}_i\}_{i=0}^{n-1}$ with distinct elements, e.g.

$$K(\boldsymbol{t}, \boldsymbol{x}) = \exp\big(-\|\boldsymbol{t} - \boldsymbol{x}\|^2/h^2\big), \quad \mathscr{X} = \mathbb{R}^d, \qquad \text{squared exponential}$$

$$K(\boldsymbol{t}, \boldsymbol{x}) = \prod_{\ell=1}^{d} \left[1 + \frac{\gamma_\ell^2}{2}\big(|t_\ell - 1/2| + |x_\ell - 1/2| - |t_\ell - x_\ell|\big)\right], \quad \mathscr{X} = [0,1]^d$$

centered discrepancy

Discrepancy from kernels

Suppose that

- $K : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$ is a symmetric, positive definite kernel,
- $\mathscr{X}$ is the sample space for random variables/vectors $\boldsymbol{X}$ and $\boldsymbol{Z}$
- $\boldsymbol{X}$ has CDF $F_{\boldsymbol{X}}$ and $\boldsymbol{Z}$ has CDF $F_{\boldsymbol{Z}}$

Then discrepancy measures the difference between two distributions:

$$D^2\big(F_{\boldsymbol{X}}, F_{\boldsymbol{Z}}; K\big) = \mathbb{E}_{\boldsymbol{X},\boldsymbol{X}' \sim F_{\boldsymbol{X}}} K(\boldsymbol{X}, \boldsymbol{X}') - 2\mathbb{E}_{\boldsymbol{X} \sim F_{\boldsymbol{X}}, \boldsymbol{Z} \sim F_{\boldsymbol{Z}}} K(\boldsymbol{X}, \boldsymbol{Z}) + \mathbb{E}_{\boldsymbol{Z},\boldsymbol{Z}' \sim F_{\boldsymbol{Z}}} K(\boldsymbol{Z}, \boldsymbol{Z}')$$

$$= \int_{\mathscr{X} \times \mathscr{X}} K(\boldsymbol{x}, \boldsymbol{x}')\, \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}') - 2 \int_{\mathscr{X} \times \mathscr{X}} K(\boldsymbol{x}, \boldsymbol{z})\, \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) \mathrm{d}F_{\boldsymbol{Z}}(\boldsymbol{z})$$

$$+ \int_{\mathscr{X} \times \mathscr{X}} K(\boldsymbol{z}, \boldsymbol{z}')\, \mathrm{d}F_{\boldsymbol{Z}}(\boldsymbol{z}) \mathrm{d}F_{\boldsymbol{Z}}(\boldsymbol{z}') \geq 0$$

Discrepancy from kernels

Then discrepancy measures the difference between two distributions:

$$
\begin{aligned}
D^2\big(F_X, F_Z; K\big) &= \int_{\mathcal{X}\times\mathcal{X}} K(\boldsymbol{x},\boldsymbol{x}')\,\mathrm{d}F_X(\boldsymbol{x})\mathrm{d}F_X(\boldsymbol{x}') - 2\int_{\mathcal{X}\times\mathcal{X}} K(\boldsymbol{x},z)\,\mathrm{d}F_X(\boldsymbol{x})\mathrm{d}F_Z(z) \\
&\qquad + \int_{\mathcal{X}\times\mathcal{X}} K(z,z')\,\mathrm{d}F_Z(z)\mathrm{d}F_Z(z') \\
&= \int_{\mathcal{X}\times\mathcal{X}} K(\boldsymbol{x},\boldsymbol{x}')\,\mathrm{d}[F_X(\boldsymbol{x}) - F_Z(\boldsymbol{x})\mathrm{d}[F_X(\boldsymbol{x}') - F_Z(\boldsymbol{x}')] \\
&= \int_{\mathcal{X}\times\mathcal{X}} K(\boldsymbol{x},\boldsymbol{x}')\,\varrho_X(\boldsymbol{x})\varrho_X(\boldsymbol{x}')\,\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{x}' - 2\int_{\mathcal{X}\times\mathcal{X}} K(\boldsymbol{x},z)\,\varrho_X(\boldsymbol{x})\varrho_Z(z)\,\mathrm{d}\boldsymbol{x}\mathrm{d}z \\
&\qquad + \int_{\mathcal{X}\times\mathcal{X}} K(z,z')\,\varrho_Z(z)\varrho_Z(z')\,\mathrm{d}z\mathrm{d}z' \geq 0
\end{aligned}
$$

Discrepancy with empirical distributions

The formula for the squared discrepancy works if one or both of the distributions is the empirical distribution of a sample.

$$D^2\left(F_X, F_{\{z_i\}_{i=0}^{n-1}}; K\right) = D^2\left(F_X, \{z_i\}_{i=0}^{n-1}; K\right)$$

$$= \int_{\mathcal{X}\times\mathcal{X}} K(\boldsymbol{x},\boldsymbol{x}')\,\mathrm{d}F_X(\boldsymbol{x})\mathrm{d}F_X(\boldsymbol{x}') - \frac{2}{n}\sum_{i=0}^{n-1}\int_{\mathcal{X}} K(\boldsymbol{x},z_i)\,\mathrm{d}F_X(\boldsymbol{x})$$

$$+\frac{1}{n^2}\sum_{i,j=0}^{n-1} K(z_i,z_j)$$

$$D^2\left(\{\boldsymbol{x}_i\}_{i=0}^{m-1}, \{z_i\}_{i=0}^{n-1}; K\right) = \frac{1}{m^2}\sum_{i,j=0}^{m-1} K(\boldsymbol{x}_i,\boldsymbol{x}_j) - \frac{2}{mn}\sum_{i,j=0}^{m-1,n-1} K(\boldsymbol{x}_i,z_j) + \frac{1}{n^2}\sum_{i,j=0}^{n-1} K(z_i,z_j)$$

Unbiased estimates of squared discrepancy

The squared discrepancy is non-negative, but we can construct unbiased and possibly negative estimates of $D^2\big(F_X, F_Z; K\big)$ based on IID samples $\{\boldsymbol{x}_i\}_{i=0}^{m-1}$ and/or $\{z_i\}_{i=0}^{n-1}$

$$D^2_{\mathrm{unb}}\Big(F_X, \{z_i\}_{i=0}^{n-1}; K\Big) = \int_{\mathcal{X}\times\mathcal{X}} K(\boldsymbol{x}, \boldsymbol{x}')\, \mathrm{d}\varrho_X(\boldsymbol{x})\mathrm{d}\varrho_X(\boldsymbol{x}') - \frac{2}{n}\sum_{i=0}^{n-1}\int_{\mathcal{X}} K(\boldsymbol{x}, z_i)\, \mathrm{d}\varrho_X(\boldsymbol{x})$$

$$+\frac{1}{n(n-1)}\sum_{\substack{i,j=0\\ i\neq j}}^{n-1} K(z_i, z_j)$$

$$D^2_{\mathrm{unb}}\Big(\{\boldsymbol{x}_i\}_{i=0}^{m-1}, \{z_i\}_{i=0}^{n-1}; K\Big) = \frac{1}{m(m-1)}\sum_{\substack{i,j=0\\ i\neq j}}^{m-1} K(\boldsymbol{x}_i, \boldsymbol{x}_i) - \frac{2}{mn}\sum_{i,j=0}^{m-1,n-1} K(\boldsymbol{x}_i, z_j)$$

$$+\frac{1}{n(n-1)}\sum_{\substack{i,j=0\\ i\neq j}}^{n-1} K(z_i, z_j)$$

Discrepancy as worst case error

Let $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a symmetric, positive definite kernel. Then

- $\mathcal{H}$:= completion (add limits of Cauchy sequences) of
$$\{c_0 K(\cdot\,, \boldsymbol{x}_0) + \cdots + c_{n-1} K(\cdot\,, \boldsymbol{x}_{n-1}) : n \in \mathbb{N},\, c_i \in \mathbb{R},\, \boldsymbol{x}_i \in \mathcal{X}\}$$
is a Hilbert space (vector space + inner product) of functions defined on $\mathcal{X}$ with reproducing kernel K

- Which means $f(\boldsymbol{x}) = \langle K(\cdot\,, \boldsymbol{x}), f\rangle_{\mathcal{H}}$ for all $f \in \mathcal{H}$, $\boldsymbol{x} \in \mathcal{X}$

- Suppose that $\int_{\mathcal{X}\times\mathcal{X}} K(\boldsymbol{x}, \boldsymbol{x}')\, \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x})\mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}')$ is defined and finite, which means that $f \mapsto \int_{\mathcal{X}} f(\boldsymbol{x})\, \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x})$ is a bounded linear functional on $\mathcal{H}$

- This means that cubature error, $f \mapsto \int_{\mathcal{X}} f(\boldsymbol{x})\, \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) - n^{-1}\sum_{i=0}^{n-1} f(\boldsymbol{z}_i)$ is also a bounded linear functional on $\mathcal{H}$

Discrepancy as worst case error (cont'd)

- Reproducing property: $f(\boldsymbol{x}) = \langle K(\,\cdot\,,\boldsymbol{x}), f\rangle_{\mathscr{H}}$ for all $f \in \mathscr{H}$, $\boldsymbol{x} \in \mathscr{X}$
- Cubature error, $f \mapsto \int_{\mathscr{X}} f(\boldsymbol{x})\,\mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) - n^{-1}\sum_{i=0}^{n-1} f(\boldsymbol{z}_i)$, is a bounded linear functional on $\mathscr{H}$
- By the Riesz Representation Theorem there exists a $\zeta \in \mathscr{H}$ such that

$$\int_{\mathscr{X}} f(\boldsymbol{x})\,\mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) - \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{z}_i) = \langle \zeta, f\rangle_{\mathscr{H}} \quad \forall f \in \mathscr{H}$$

- The Cauchy-Schwarz inequality separates the contribution to the error from $\{\boldsymbol{z}_i\}_{i=0}^{n-1}$ and from f

$$\left|\int_{\mathscr{X}} f(\boldsymbol{x})\,\mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) - \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{z}_i)\right| = \left|\langle \zeta, f\rangle_{\mathscr{H}}\right| \le \|\zeta\|_{\mathscr{H}}\|f\|_{\mathscr{H}} \quad \forall f \in \mathscr{H}$$

Discrepancy as worst case error (cont'd)

$$\left| \int_{\mathscr{X}} f(\boldsymbol{x})\,\mathrm{d}F_X(\boldsymbol{x}) - \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{z}_i) \right| = \left| \langle \zeta, f \rangle_{\mathscr{H}} \right| \leq \|\zeta\|_{\mathscr{H}} \|f\|_{\mathscr{H}} \quad \forall f \in \mathscr{H}$$

- By the reproducing property and the Riesz Representation Theorem,

$$\zeta(\boldsymbol{x}') = \langle K(\,\cdot\,, \boldsymbol{x}'), \zeta \rangle_{\mathscr{H}} = \int_{\mathscr{X}} K(\boldsymbol{x}, \boldsymbol{x}')\,\mathrm{d}F_X(\boldsymbol{x}) - \frac{1}{n}\sum_{i=0}^{n-1} K(\boldsymbol{z}_i, \boldsymbol{x}')$$

$$\|\zeta\|_{\mathscr{H}}^2 = \langle \zeta, \zeta \rangle_{\mathscr{H}} = \int_{\mathscr{X}\times\mathscr{X}} K(\boldsymbol{x}, \boldsymbol{x}')\,\mathrm{d}F_X(\boldsymbol{x})\mathrm{d}F_X(\boldsymbol{x}') - \frac{2}{n}\sum_{i=0}^{n-1} \int_{\mathscr{X}} K(\boldsymbol{x}, \boldsymbol{z}_i)\,\mathrm{d}F_X(\boldsymbol{x})$$

$$+ \frac{1}{n^2}\sum_{i,j=0}^{n-1} K(\boldsymbol{z}_i, \boldsymbol{z}_j) = D^2\left(F_X, \{\boldsymbol{z}_i\}_{i=0}^{n-1}; K\right)$$

Discrepancy as worst case error (cont'd)

In particular, for uniform target distributions

$$\left| \int_{[0,1]^d \times [0,1]^d} f(\boldsymbol{x})\,\mathrm{d}(\boldsymbol{x}) - \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{z}_i) \right| \leq D\left(\mathscr{U}[0,1]^d, \{\boldsymbol{z}_i\}_{i=0}^{n-1}; K\right) \|f\|_{\mathscr{H}} \quad \forall f \in \mathscr{H}$$

where

$$D^2\left(\mathscr{U}[0,1]^d, \{\boldsymbol{z}_i\}_{i=0}^{n-1}; K\right) = \int_{[0,1]^d \times [0,1]^d} K(\boldsymbol{x}, \boldsymbol{x}')\,\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{x}' - \frac{2}{n}\sum_{i=0}^{n-1} \int_{[0,1]^d} K(\boldsymbol{x}, \boldsymbol{z}_i)\,\mathrm{d}\boldsymbol{x}$$

$$+ \frac{1}{n^2}\sum_{i,j=0}^{n-1} K(\boldsymbol{z}_i, \boldsymbol{z}_j)$$

After Test 1, how do we improve?

- Go to menti.com
- Use code 8510 7177
- Two slides
  - One to help me know how I can help
  - One for you to reflect on what you will do differently

My response to your feedback

- You requested more worked examples, especially those that mimic test problems
  - You need to study to learn concepts, not to be able to do problems
  - I will do more examples in class; you are welcome to ask for more
  - I will assign more problems for homework of the type that may appear on tests (see the revised Assignment 3)
  - I encourage you to look at old tests
  - My tests will not intentionally copy class examples or old test problems

My response to your feedback

- You asked for more background material and were concerned that the class required advanced mathematics or statistics.
  - The statistics and mathematics required on the tests are a calculus and a calculus-level probability class
    - How to take expectations
    - The binomial, exponential, and normal/Gaussian distributions
    - Sample means and variances
    - Confidence intervals via the Central Limit Theorem
  - I do not expect you to know the details of functional analysis or stochastic processes, but to understand the overall argument
- To help me understand what advanced mathematics or statistics you need help with please respond at mentimeter.com 2644 2309

My response to your feedback

- There were comments about the test format, how to recover from a low score and number of tests
  - In-person tests help me gauge what you have learned
  - I will take the highest of your two test scores rather than average the two
  - This is a graduate class suitable for masters and doctoral students
- Please provide further and continuing feedback through
  - Comments in class
  - Emails
  - Future Mentimeter
  - Midterm course evaluation from the Center for Learning Innovation

Discrepancy as average case error

Suppose that

- f is drawn from a Gaussian process with zero mean and covariance kernel $K : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$, i.e., $f \sim \mathscr{GP}(0,K)$.
- Do not need to specify the sample space for f, which will almost surely have less smoothness than the Hilbert space with reproducing kernel K.

The mean squared error for a deterministic cubature rule is

$$\mathbb{E}_{f \in \mathscr{GP}(0,K)} \left| \int_{\mathcal{X}} f(\boldsymbol{x}) \, \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\boldsymbol{z}_i) \right|^2$$

$$= \int_{\mathscr{X} \times \mathscr{X}} \mathbb{E}_{f \in \mathscr{GP}(0,K)} \left[f(\boldsymbol{x}) f(\boldsymbol{x}') \right] \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathscr{X}} \mathbb{E}_{f \in \mathscr{GP}(0,K)} \left[f(\boldsymbol{x}) f(\boldsymbol{z}_i) \right] \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x})$$

$$+ \frac{1}{n^2} \sum_{i,j=0}^{n-1} \mathbb{E}_{f \in \mathscr{GP}(0,K)} \left[f(\boldsymbol{z}_i) f(\boldsymbol{z}_j) \right]$$

Discrepancy as average case error (cont'd)

The mean squared error for a deterministic cubature rule is

$$\mathbb{E}_{f \in \mathscr{GP}(0,K)} \left| \int_{\mathcal{X}} f(\boldsymbol{x}) \, \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) - \frac{1}{n} \sum_{i=0}^{n-1} f(\boldsymbol{z}_i) \right|^2$$

$$= \int_{\mathscr{X} \times \mathscr{X}} \mathbb{E}_{f \in \mathscr{GP}(0,K)} \left[f(\boldsymbol{x}) f(\boldsymbol{x}') \right] \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathscr{X}} \mathbb{E}_{f \in \mathscr{GP}(0,K)} \left[f(\boldsymbol{x}) f(\boldsymbol{z}_i) \right] \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x})$$

$$+ \frac{1}{n^2} \sum_{i,j=0}^{n-1} \mathbb{E}_{f \in \mathscr{GP}(0,K)} \left[f(\boldsymbol{z}_i) f(\boldsymbol{z}_j) \right]$$

$$= \int_{\mathscr{X} \times \mathscr{X}} K(\boldsymbol{x}, \boldsymbol{x}') \, \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}') - \frac{2}{n} \sum_{i=0}^{n-1} \int_{\mathscr{X}} K(\boldsymbol{x}, \boldsymbol{z}_i) \, \mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) + \frac{1}{n^2} \sum_{i,j=0}^{n-1} K(\boldsymbol{z}_i, (\boldsymbol{z}_j)$$

$$= D^2 \left(F_{\boldsymbol{X}}, \{\boldsymbol{z}_i\}_{i=0}^{n-1} \right)$$

Observations about discrepancy

- The value of the discrepancy depends on the choice of kernel, K, and the parameters that define it
  - K may be chosen for convenience
  - Should choose the hyperparameters of K to ensure that your f is in the middle; this can be done by empirical Bayes/maximum likelihood estimation in the average case setting
  - K may be chosen based on knowledge about properties of f, such as domain, smoothness, periodicity, importance of different coordinates
  - $D(\cdot, \cdot; c^2 K) = |c| D(\cdot, \cdot; K)$
- $D(F_X, \{z\}_{i=0}^{n-1}; K)$ measures the quality of $\{z\}_{i=0}^{n-1}$ for estimating the mean, $\mu = \mathbb{E}[f(X)]$, by the sample mean $\hat{\mu}_n = n^{-1} \sum_{i=0}^{n-1} f(z)$ for
  - f in a Hilbert space, $\mathscr{H}$, with reproducing kernel K (don't need to know $\|\cdot\|_{\mathscr{H}}$ explicitly) or
  - $f \sim \mathscr{GP}(0, K)$

Observations about discrepancy (cont'd)

- The mean square discrepancy of an IID random sample is

$$\mathbb{E}\left[D^2\left(F_{\boldsymbol{X}}, \{\boldsymbol{z}_i \overset{\text{IID}}{\sim} F_{\boldsymbol{X}}\}_{i=0}^{n-1}; K\right)\right] = \frac{1}{n}\left[\int_{\mathcal{X}} K(\boldsymbol{x},\boldsymbol{x})\,\mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) - \int_{\mathcal{X}\times\mathcal{X}} K(\boldsymbol{x},\boldsymbol{x}')\,\mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x})\mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}')\right]$$

- The deterministic cubature error bound cannot be used constructively to bound the error because the norm of f cannot be estimated

- If $f \sim \mathscr{GP}(0,K)$, and the parameters of K are estimated by empirical Bayes (maximum likelihood), then one can construct credible intervals for the cubature error

$$\mathbb{P}_{f\in\mathscr{GP}(0,K)}\left[\left|\int_{\mathcal{X}} f(\boldsymbol{x})\,\mathrm{d}F_{\boldsymbol{X}}(\boldsymbol{x}) - \frac{1}{n}\sum_{i=0}^{n-1} f(\boldsymbol{z}_i)\right| \le \frac{2.58D\left(F_{\boldsymbol{X}}, \{\boldsymbol{z}_i\}_{i=0}^{n-1}; K\right)}{\sqrt{n}}\right] \ge 99\,\%$$

Example of the Centered discrepancy

$$K(\boldsymbol{t},\boldsymbol{x}) = \prod_{\ell=1}^{d}\left[1+\frac{\gamma_\ell^2}{2}\big(|t_\ell - 1/2| + |x_\ell - 1/2| - |t_\ell - x_\ell|\big)\right], \quad \mathcal{X} = [0,1]^d$$

$$\|f\|_{\mathcal{H}}^2 = \|f(0.5,\ldots,0.5)\|_2^2$$

$$+ \left\| \frac{\partial f(x_1,0.5,\ldots,0.5)}{\gamma_1\, \partial x_1} \right\|_2^2 + \left\| \frac{\partial f(0.5,x_2,0.5,\ldots,0.5)}{\gamma_2\, \partial x_2} \right\|_2^2 + \cdots$$

$$+ \left\| \frac{\partial f(x_1,x_2,0.5,\ldots,0.5)}{\gamma_1\gamma_2\, \partial x_1 \partial x_2} \right\|_2^2 + \left\| \frac{\partial f(x_1,0.5,x_3,0.5,\ldots,0.5)}{\gamma_1\gamma_3\, \partial x_1 \partial x_3} \right\|_2^2 + \cdots$$

$$+ \ldots + \left\| \frac{\partial f(x_1,\ldots,x_d)}{\gamma_1\cdots\gamma_d\, \partial x_1\cdots\partial x_d} \right\|_2^2$$

And now back to MCMC

Overcoming challenges of MCMC

- The probability densities where we use MCMC are typically spiky and often multimodal
  - Small step sizes may give samples that do not represent the distribution well
  - Large step sizes lead to low acceptance rates
- One may try pilot sampling or multiple chains
- Parallel tempering (replica exchange) combines benefits of large step sizes with multiple chains

Maximum likelihood estimation (MLE)

Let $\boldsymbol{\theta}$ be a parameter to be estimated, and let

$$\begin{aligned}\varrho_{\text{DATA}}(\text{data}|\boldsymbol{\theta}) &= \text{density for random DATA given parameter } \boldsymbol{\theta} \\ &= \text{also known as likelihood}(\boldsymbol{\theta}|\text{data})\end{aligned}$$

$\boldsymbol{\theta}_{\text{MLE}}$ maximizes likelihood($\boldsymbol{\theta}$ | data) is called the maximum likelihood estimate

E.g. if the data is $y_0, \ldots, y_{n-1} \overset{\text{IID}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then

$$\varrho_{\text{DATA}}\left(\{y_i\}_{i=0}^{n-1} \,|\, (\mu, \sigma^2)\right) = \frac{1}{(2\pi\sigma)^{n/2}} \exp\left(-\sum_{i=0}^{n-1} (y_i - \mu)^2\right)$$

$$\mu_{\text{MLE}} = \frac{1}{n}\sum_{i=0}^{n-1} y_i, \qquad \sigma^2_{\text{MLE}} = \frac{1}{n}\sum_{i=0}^{n-1} (y_i - \mu_{\text{MLE}})^2$$

Bayesian Inference

$$\begin{aligned}
\varrho_{\text{prior}}(\boldsymbol{\theta}) &= \text{density expressing prior belief about parameter } \boldsymbol{\Theta} \text{ at } \boldsymbol{\theta} \\
\varrho_{\text{DATA}|\boldsymbol{\Theta}}(\text{data}|\boldsymbol{\theta}) &= \text{density for DATA given } \boldsymbol{\Theta} = \boldsymbol{\theta} \text{ evaluated at data} \\
&= \text{also know as likelihood}_{\boldsymbol{\Theta}|\text{DATA}}(\boldsymbol{\theta}|\text{data})
\end{aligned}$$

Then

$$\begin{aligned}
\varrho_{\text{posterior}}(\boldsymbol{\theta}|\text{data}) &= \text{density expressing belief about } \boldsymbol{\Theta} \text{ given data and prior belief} \\
&= \frac{\varrho_{\text{joint}}(\boldsymbol{\theta},\text{data})}{\varrho_{\text{DATA}}(\text{data})} = \frac{\varrho_{\text{DATA}|\boldsymbol{\Theta}}(\text{data}|\boldsymbol{\theta})\,\varrho_{\text{prior}}(\boldsymbol{\theta})}{\varrho_{\text{DATA}}(\text{data})} \\
&= \frac{\text{likelihood}_{\boldsymbol{\Theta}|\text{DATA}}(\boldsymbol{\theta}|\text{data})\,\varrho_{\text{prior}}(\boldsymbol{\theta})}{\varrho_{\text{DATA}}(\text{data})} \propto \text{likelihood}_{\boldsymbol{\Theta}|\text{DATA}}(\boldsymbol{\theta}|\text{data})\,\varrho_{\text{prior}}(\boldsymbol{\theta})
\end{aligned}$$

Have unnormalized posterior, i.e. can ignore anything depending only on data

May compute maximum a posteriori (MAP) or posterior mean of $\boldsymbol{\theta}$ through MCMC sampling

Queues

Imagine a customer service system with random arrival times of customers and random service times. There may be several stages. We consider the case where i represents an event (not a fixed time increment) E.g.,

$$A_i = \text{arrival time of next customer} \overset{\text{IID}}{\sim} F_A$$
$$N_i = \text{number of customers in the queue}$$
$$S_i = \text{service time of first customer in queue} \overset{\text{IID}}{\sim} F_S$$

```
Given A[0] ~ F_A, N[0]←0, S[0] ~ F_S, n #total number of customers
While i < n
  If N[i] = 0  #new customer
    N[i+1]←1, A[i+1] ~ F_A, S[i+1]←S[i]
  ElseIf A[i] < S[i]  #new customer
    N[i+1]←N[i+1]+1, A[i+1] ~ F_A, S[i+1]←S[i]−A[i]
  Else  #finish serving customer
    N[i+1]←N[i+1]−1, A[i+1] = A[i]−S[i], S[I+1] ~ F_S
  i←i+1
```

Queues

Imagine a customer service system with random arrival times of customers and random service times. There may be several stages. Suppose that you want to measure customer waiting time and total time

$$A_i = \text{arrival time of next customer} \overset{\text{IID}}{\sim} F_A$$

$$N_i = \text{number of customers in the queue}$$

$$S_i = \text{service time of first customer in queue} \overset{\text{IID}}{\sim} F_S$$

$$T_i = \text{total time passed}$$

$$W_j = \text{waiting time plus service time for the } j^{\text{th}} \text{ customer}$$

Queues

```
Given A[0] ~ F_A, N[0]←0, S[0] ~ F_S, T[0]←0
n #total number of customers
time #maximum time
F←0 #index of first customer in queue
E←0 #index of next customer to enter queue
While i < n and T[I] < time
  If N[i] = 0  #get new customer
    N[i+1]←1, A[i+1] ~ F_A, S[i+1]←S[i], T[i+1]←T[i+1]+A[i]
    W[E]←A[i], E←E+1
  ElseIf A[i] < S[i]  #get new customer
    N[i+1]←N[i+1]+1, A[i+1] ~ F_A, S[i+1]←S[i]−A[i], T[i+1]←T[i+1]+A[i]
    W[E]←A[i], W[E−1]←W[E−1]+A[i], …, W[F]←W[F]+A[i], E←E+1
  Else  #finish serving customer
    N[i+1]←N[i+1]−1, A[i+1] = A[I]−S[i], S[i+1] ~ F_S, T[i+1]←T[i+1]+S[i]
    W[E−1]←W[E−1]+S[i], …, W[F]←W[F]+S[i], F←F+1
  i←i+1
```